1 Problem setup, identification and inference

1.1 Setup

A *response set* is a finite collection $U = \{1, \dots, N\}$ of records. Record i carries an outcome y_i, covariates x_i that include a detector score with flag $d_i \in \{0, 1\}$, and an origin a_i, the share of the response that is machine-written ($a_i \in \{0, 1\}$ in the main text, $a_i \in [0, 1]$ for hybrid authorship). Write $h_i = 1 - a_i$, $N_H = \sum_U h_i > 0$ and $N_A = N - N_H$, $\varepsilon = N_A/N$. The estimand is the human mean of the response set,

$$\mu_H = \frac{\sum_{i \in U} h_i \, y_i}{\sum_{i \in U} h_i}, \tag{1}$$

and μ_A is the corresponding machine mean. Outcomes, covariates and flags are observed for every record; origins are not. An *audit* verifies the origin of a subset $S \subset U$ of m records.

Assumption 1 (D1, probability audit)**.** S is drawn by a randomised design chosen by the analyst, with inclusion probabilities $\rho_i = \Pr(i \in S) > 0$ known for every audited record. For a design with several stages the probabilities are conditional on the earlier stages (Lemma 2).

Assumption 2 (D2, correct verification)**.** For $i \in S$ the audit returns a_i. (Proposition 5 relaxes this.)

Assumption 3 (D3, complete first phase)**.** (y_i, x_i, d_i) is observed for every $i \in U$.

Assumption 4 (M1, non-differential flag)**.** Within each origin class the flag is uncorrelated with the outcome: $\mathrm{Cov}(D, Y \mid A = a) = 0$ for $a \in \{0, 1\}$.

D1–D3 are properties of the audit that the analyst controls. M1 is a property of the detector and the outcome that cannot be checked without verified origins; it was the identifying condition of the first version of this paper. It is needed below only for the moment inversion (Proposition 1); the estimators we recommend do not use it. Probabilities and moments written with capital letters (Y, A, D) refer to any joint distribution of outcome, origin and flag, in particular the empirical distribution of the response set, so that the identities of Section 1.2 hold exactly on every response set. Write $\bar{f} = \Pr(D{=}1 \mid A{=}0)$ and $\bar{s} = \Pr(D{=}1 \mid A{=}1)$ for the false-positive rate and the sensitivity, and $C_a = \mathrm{Cov}(D, Y \mid A = a)$.

1.2 Why deletion and moment inversion are biased

The naive mean has bias $\varepsilon(\mu_A - \mu_H)$. Deleting flagged records gives $\mathbb{E}[Y \mid D{=}0]$.

Lemma 1 (Deletion bias)**.** *Let* $\bar{f} < 1$ *and* $\varepsilon_{\mathrm{res}} = \Pr(A{=}1 \mid D{=}0) = \varepsilon(1 - \bar{s})/\{(1 - \varepsilon)(1 - \bar{f}) + \varepsilon(1 - \bar{s})\}$. *Then*

$$\mathbb{E}[Y \mid D{=}0] - \mu_H = \varepsilon_{\mathrm{res}} \, \Delta_{\mathrm{surv}} - \frac{C_0}{1 - \bar{f}}, \qquad \Delta_{\mathrm{surv}} = \mathbb{E}[Y \mid A{=}1, D{=}0] - \mathbb{E}[Y \mid A{=}0, D{=}0]. \tag{2}$$

Proof. By total probability over the origin of the unflagged records, $\mathbb{E}[Y \mid D{=}0] = \mathbb{E}[Y \mid A{=}0, D{=}0] + \varepsilon_{\mathrm{res}}\Delta_{\mathrm{surv}}$. Among humans, $\mathbb{E}[Y(1 - D) \mid A{=}0] = \mu_H(1 - \bar{f}) - C_0$, so $\mathbb{E}[Y \mid A{=}0, D{=}0] = \mu_H - C_0/(1 - \bar{f})$. □

The first term is residual contamination: machine records that survive deletion, times their outcome gap to the surviving humans. The second is a selection effect on the humans themselves: if falsely flagged people answer differently from other people ($C_0 \neq 0$), deleting them moves the mean even when the detector catches every machine record. With a style stratum G and $f(G) = \Pr(D{=}1 \mid A{=}0, G)$, $C_0 = \mathrm{Cov}(f(G), \mathbb{E}[Y \mid A{=}0, G]) + \mathbb{E}[\mathrm{Cov}(D, Y \mid A{=}0, G)]$; the first submission assumed the last term away.

Proposition 1 (Moment inversion)**.** *Let $0 < \varepsilon < 1$ and $\bar{s} \neq \bar{f}$, with $(\bar{s}, \bar{f}, \varepsilon)$ known. The solution in (μ_H, μ_A) of*

$$\mathbb{E}[YD] = (1-\varepsilon)\bar{f}\,\mu_H + \varepsilon\bar{s}\,\mu_A, \qquad \mathbb{E}[Y(1-D)] = (1-\varepsilon)(1-\bar{f})\,\mu_H + \varepsilon(1-\bar{s})\,\mu_A$$

has first component

$$\tilde{\mu}_H = \mu_H - \frac{C_0 + \frac{\varepsilon}{1-\varepsilon} C_1}{\bar{s} - \bar{f}}. \tag{3}$$

Hence $\tilde{\mu}_H = \mu_H$ under M1, and without M1 the inversion is biased by the second term, which grows without bound as $\bar{s} - \bar{f} \to 0$.

Proof. The determinant of the system is $\varepsilon(1-\varepsilon)(\bar{f} - \bar{s}) \neq 0$ and Cramer's rule gives $\tilde{\mu}_H = \{(1-\bar{s})\mathbb{E}[YD] - \bar{s}\,\mathbb{E}[Y(1-D)]\}/\{(1-\varepsilon)(\bar{f}-\bar{s})\}$. In truth $\mathbb{E}[YD] = (1-\varepsilon)(\bar{f}\mu_H + C_0) + \varepsilon(\bar{s}\mu_A + C_1)$ and $\mathbb{E}[Y(1-D)] = (1-\varepsilon)\{(1-\bar{f})\mu_H - C_0\} + \varepsilon\{(1-\bar{s})\mu_A - C_1\}$. Substituting, the terms in μ_A cancel and the numerator equals $(1-\varepsilon)\{(\bar{f} - \bar{s})\mu_H + C_0\} + \varepsilon C_1$. □

Equation (3) is the analogue of a weak-instrument bias: the violation of M1 is divided by the detector's separation, so a weak detector amplifies it.

1.3 Design-based identification

For a function g on U write $t_g = \sum_U g_i$ and $\hat{t}_g = \sum_S g_i/\rho_i$ for its Horvitz–Thompson estimator; under D1, $\mathbb{E}_p[\hat{t}_g] = t_g$, where $\mathbb{E}_p$ is expectation over the audit design with the response set held fixed.

Definition 1 (Difference-ratio family)**.** Let $\eta : U \to \mathbb{R}$ be a predictor of the origin, computed from (x_i, d_i), and $\lambda \in \mathbb{R}$. Put

$$\hat{N}_\lambda = \lambda \sum_{i \in U} y_i(1-\eta_i) + \sum_{i \in S} \frac{y_i\{h_i - \lambda(1-\eta_i)\}}{\rho_i}, \qquad \hat{D}_\lambda = \lambda \sum_{i \in U}(1-\eta_i) + \sum_{i \in S} \frac{h_i - \lambda(1-\eta_i)}{\rho_i}, \tag{4}$$

and $\hat{\mu} = \hat{N}_\lambda / \hat{D}_\lambda$.

The family contains the two-phase Hájek estimator ($\lambda = 0$), flag-PPI ($\eta_i = d_i$, $\lambda = 1$), RC-PPI (η_i a fitted probability that record i is machine-written, $\lambda = 1$) and their power-tuned versions (flag-PPI++, RC-PPI++; a data-chosen λ).

Theorem 1 (Identification without M1)**.** *Assume D1–D3 and let η and λ not depend on S. Then*

1. *$\mathbb{E}_p[\hat{N}_\lambda] = N_H\,\mu_H$ and $\mathbb{E}_p[\hat{D}_\lambda] = N_H$;*
2. *on the event $\hat{D}_\lambda \neq 0$,*

$$\hat{\mu} - \mu_H = \frac{\hat{t}_e - t_e}{\hat{D}_\lambda}, \qquad e_i = \{h_i - \lambda(1-\eta_i)\}(y_i - \mu_H). \tag{5}$$

No assumption links the predictor, the flag or the outcome: M1 is not used and η may be arbitrarily wrong.

Proof. (1) The sums over S are Horvitz–Thompson estimators of $\sum_U y_i\{h_i - \lambda(1-\eta_i)\}$ and $\sum_U\{h_i - \lambda(1-\eta_i)\}$; adding the first-phase sums leaves $\sum_U y_i h_i = N_H\mu_H$ and $\sum_U h_i = N_H$. (2) $\hat{N}_\lambda - \mu_H\hat{D}_\lambda = \lambda\sum_U(y_i - \mu_H)(1-\eta_i) + \hat{t}_e$, and $t_e = \sum_U h_i(y_i - \mu_H) - \lambda\sum_U(1-\eta_i)(y_i - \mu_H) = -\lambda\sum_U(1-\eta_i)(y_i - \mu_H)$ because $\sum_U h_i(y_i - \mu_H) = 0$ by (1). □

The human mean is thus identified by the audit design alone. The detector enters only through the residual e_i: it cannot bias the estimate, only change its variance. For $\lambda = 1$ the residual is $(\eta_i - a_i)(y_i - \mu_H)$, the prediction error times the outcome deviation, which vanishes for a perfect predictor. All statements hold verbatim for graded origins $a_i \in [0, 1]$, since (4) is linear in h_i.

1.4 Variance, intervals and efficiency

Theorem 2 (Linearised variance and normal limit)**.** *Assume D1–D3, and consider a sequence of response sets and designs along which (i) N_H/N is bounded away from zero and y and η are bounded; (ii) Horvitz–Thompson totals of bounded variables satisfy a central limit theorem under the design and their design variance has a consistent estimator $\hat{V}(\cdot)$ (for stratified simple random sampling without replacement, the conditions of Bickel and Freedman, 1984); (iii) the fitted predictor and the tuning constant are stable: there are fixed η^*, λ^* with $N^{-1}\sum_U(\hat{\eta}_i - \eta_i^*)^2 \to 0$ and $\hat{\lambda} \to \lambda^*$ in probability, and the predictor applied to an audited record is fitted without that record (cross-fitting). Then $(\hat{\mu} - \mu_H)/\widehat{\mathrm{se}} \to \mathcal{N}(0,1)$ in distribution, with*

$$\widehat{\mathrm{se}}^{\,2} = \frac{\hat{V}(\hat{e})}{\hat{D}_\lambda^2}, \qquad \hat{e}_i = \{h_i - \hat{\lambda}(1-\hat{\eta}_i)\}(y_i - \hat{\mu}), \tag{6}$$

where for stratified simple random sampling $\hat{V}(\hat{e}) = \sum_h N_h^2(1 - m_h/N_h)\, s^2_{\hat{e},h}/m_h$.

Proof sketch. For fixed (η, λ), (5) is exact, $\hat{D}_\lambda/N_H \to 1$ in probability by (i)–(ii), and the result follows from the central limit theorem for $\hat{t}_e$ and Slutsky's lemma; replacing μ_H by $\hat{\mu}$ in the residual changes $\hat{V}$ by a factor $1 + o_p(1)$. For a fitted predictor, write $\hat{N}_\lambda - \mu_H \hat{D}_\lambda = \hat{t}_u - \lambda\{\hat{t}_{v(\bar{\eta})} - t_{v(\bar{\eta})}\}$ with $u_i = h_i(y_i - \mu_H)$, $v(\eta)_i = (1 - \eta_i)(y_i - \mu_H)$, $\tilde{\eta}$ the out-of-fold predictions on the audit and $\bar{\eta}$ the predictions on U. The first term does not involve the predictor. By (iii) the second differs from $\hat{t}_{v(\eta^*)} - t_{v(\eta^*)}$ by $o_p(Nm^{-1/2})$, the argument of cross-prediction-powered inference (Zrnic and Candès, 2024) and of model-assisted survey estimation with estimated working models (Breidt and Opsomer, 2017); the derivative of the error in λ is itself $O_p(Nm^{-1/2})$, so a consistent $\hat{\lambda}$ may be plugged in (Angelopoulos et al., 2023). □

Theorem 2 is asymptotic. With 18 to 90 audited records the intervals we report add two finite-sample devices (Remark 1), and their coverage is an empirical question, answered by endpoint PE2.

Corollary 1 (Power tuning never loses to the audit-only estimator)**.** *With u, v as above, the design variance of $\hat{t}_u - \lambda\hat{t}_v$ is $V_u - 2\lambda C_{uv} + \lambda^2 V_v$, minimised at $\lambda^* = C_{uv}/V_v$ with value $V_u(1 - \varrho_{uv}^2)$. For λ^* clipped to $[0,1]$ the value is still at most V_u, the variance at $\lambda = 0$. Hence, for any predictor, a power-tuned member of the family has asymptotic variance no larger than the two-phase Hájek estimator.*

Proof. A convex quadratic in λ restricted to $[0,1]$ attains its minimum at the clipped vertex, and $0 \in [0,1]$. □

No such ordering holds between RC-PPI++ and flag-PPI, which use different predictors; their comparison is empirical (endpoint PE1).

Remark 1 (Finite-sample interval)**.** The residuals of a good predictor are sparse: $\hat{e}_i$ is close to zero unless the predictor errs on record i. In a stratum whose few audited records are all predicted correctly, the sample variance $s^2_{\hat{e},h}$ is near zero although the stratum may contain errors. We therefore (a) shrink each stratum variance towards the variance anticipated under a working model in which the origins of the stratum are independent Bernoulli(p_h) variables: writing $\hat{e}_i = z_i^0 - a_i\delta_i$ with $\delta_i = y_i - \hat{\mu}$,

$$\mathbb{E}_\xi\big[s^2_{\hat{e},h}\big] = p_h(1-p_h)\,\overline{\delta^2}_h + s^2_{z^0 - p_h\delta,\,h}, \tag{7}$$

evaluated at the Jeffreys estimate $p_h = (k_h + \tfrac{1}{2})/(m_h + 1)$ from the k_h machine records among the m_h audited, and combined as $\{(m_h - 1)s^2_{\hat{e},h} + \kappa\,\mathbb{E}_\xi[s^2_{\hat{e},h}]\}/(m_h - 1 + \kappa)$ with $\kappa = 1$; and (b) use a Student quantile with Satterthwaite degrees of freedom $(\sum_h v_h)^2 \big/ \sum_h\{v_h^2/(m_h - 1)\}$, with v_h the stratum contributions to $\hat{V}$. Identity (7) is exact: for independent a_i with mean p, $\mathbb{E}\sum_i(\hat{e}_i - \bar{\hat{e}})^2 = (1 - \frac{1}{m})\sum_i \delta_i^2 p(1-p) + \sum_i(c_i - \bar{c})^2$ with $c_i = z_i^0 - p\delta_i$. The rule was selected on development data and is not claimed to be optimal.

1.5 Audit design

RC-PPI needs audited records of both classes to fit its predictor. Let k be the number of audited machine records.

Proposition 2 (Audit starvation)**.** *Under simple random sampling of m records, k is hypergeometric with parameters (N, N_A, m) and mean $m\varepsilon$. Under a stratified design with m_h records from a stratum with machine share ε_h, $\mathbb{E}[k] = \sum_h m_h\varepsilon_h$, which equals $m\varepsilon$ under proportional allocation.*

For $N = 900$, $m = 45$ and $\varepsilon = 0.05$, $\Pr(k \geq 4) = 0.18$ and $\Pr(k \geq 6) = 0.02$: at low contamination a small audit rarely contains enough machine records, whatever estimator is used afterwards. Only oversampling high-score strata raises $\mathbb{E}[k]$, and the next result shows what that costs.

Proposition 3 (Allocation)**.** *Under stratified simple random sampling with $\sum_h m_h = m$, ignoring integer constraints and $m_h \leq N_h$, the design variance of $\hat{t}_e$ is $\sum_h N_h^2 S_h^2/m_h - \sum_h N_h S_h^2$, where S_h^2 is the variance of e in stratum h. It is minimised by $m_h \propto N_h S_h$, with minimum $(\sum_h N_h S_h)^2/m - \sum_h N_h S_h^2$; proportional allocation exceeds the minimum by $(N/m)\sum_h N_h(S_h - \bar{S})^2$ with $\bar{S} = \sum_h N_h S_h/N$.*

Proof. By the Cauchy–Schwarz inequality $(\sum_h N_h S_h)^2 \leq (\sum_h N_h^2 S_h^2/m_h)(\sum_h m_h)$, with equality if and only if $m_h \propto N_h S_h$. Under proportional allocation the first term is $(N/m)\sum_h N_h S_h^2$, and $\sum_h N_h S_h^2 - (\sum_h N_h S_h)^2/N = \sum_h N_h(S_h - \bar{S})^2$ (Cochran, 1977, Section 5.5). □

The optimal allocation follows the residual, not the score. Under the working model of Remark 1 with $\lambda = 1$ and a calibrated predictor, $S_h^2 \approx \overline{\{\eta(1-\eta)(y-\mu_H)^2\}}_h$: the audit should go where the origin is uncertain and the outcome deviates, and the top score decile, where $\eta \approx 1$, is not such a place once the detector is informative. S_h depends on unverified origins, so an allocation close to Neyman's must be learned from a pilot audit, and then the following applies.

Lemma 2 (Two-stage audits)**.** *Let a pilot S_1 be drawn by stratified simple random sampling with sizes n_{1h}, let the second-stage sizes m_{2h} be any function of S_1 and of the origins verified in it, and let S_2 be a simple random sample of $m_{2h} \geq 1$ of the $N_h - n_{1h}$ records of stratum h outside S_1. For any g,*

$$\hat{t}_g^{(2)} = \sum_{i \in S_1} g_i + \sum_h \frac{N_h - n_{1h}}{m_{2h}} \sum_{i \in S_2 \cap h} g_i \tag{8}$$

satisfies $\mathbb{E}[\hat{t}_g^{(2)} \mid S_1] = t_g$. Consequently $\hat{t}_g^{(2)}$ is unbiased, $\mathrm{Var}(\hat{t}_g^{(2)}) = \mathbb{E}[\mathrm{Var}(\hat{t}_g^{(2)} \mid S_1)]$, and the usual stratified variance estimator applied to the second stage alone is unbiased for it when $m_{2h} \geq 2$. The single-stage weights $N_h/(n_{1h} + m_{2h})$ applied to $S_1 \cup S_2$ are in general biased when m_{2h} depends on the pilot origins.

Proof. Given S_1, the sizes m_{2h} are fixed and the second sum is the stratified expansion estimator of $\sum_{i \notin S_1} g_i$ on the population $U \setminus S_1$. Since the conditional mean is the constant t_g, the variance of the conditional mean is zero. For the last claim take one stratum with $N = 3$, $g = a = (1, 0, 0)$, $n_1 = 1$, and $m_2 = 1$ if the pilot record is machine-written and $m_2 = 2$ otherwise: the single-stage estimator has expectation $\frac{1}{3} \cdot \frac{3}{2} \cdot 1 + \frac{2}{3} \cdot 1 \cdot 1 = \frac{7}{6} \neq 1$. □

Stage-wise weights have a price: the non-pilot part of each stratum is estimated from the second-stage records only, so a stratum left with two second-stage records has a very variable expansion. This is why, at the budgets studied here, a label-adaptive design does not outperform proportional allocation, whose weights are known in advance and whose loss relative to Neyman allocation is the between-stratum spread of S_h (Proposition 3).

1.6 Certification, verification error, and regret

Let $b = \bar{y}_{d=0} - \mu_H$ be the deletion bias on the response set and $\hat{b} = \bar{y}_{d=0} - \hat{\mu}$ its estimate from an audit-based $\hat{\mu}$ with standard error $\widehat{\mathrm{se}}$. Since $\bar{y}_{d=0}$ is computed from the whole response set, $\hat{b} - b = -(\hat{\mu} - \mu_H)$.

Proposition 4 (Certifying deletion)**.** *Fix a tolerance $\tau > 0$ and certify deletion when $|\hat{b}| + q_{1-\alpha}\,\widehat{\mathrm{se}} < \tau$, with $q_{1-\alpha}$ at least the standard normal $(1-\alpha)$ quantile. Under the conclusion of Theorem 2,* $\limsup \Pr(\textit{certify}) \le \alpha$ *whenever $|b| \ge \tau$.*

Proof. If $b \ge \tau$, certification implies $\hat{b} + q_{1-\alpha}\widehat{\mathrm{se}} < \tau \le b$, that is $(\hat{\mu} - \mu_H)/\widehat{\mathrm{se}} > q_{1-\alpha}$, an event of limiting probability at most α. The case $b \le -\tau$ is symmetric. □

This is the two one-sided tests procedure for equivalence. The tolerance is a substantive choice; we use $\tau = 0.2$ standard deviations of the outcome. The rule of the first submission compared a bootstrap interval of $\hat{b}$ with zero and had no such guarantee.

Proposition 5 (Verification error)**.** *Suppose the audit returns $\tilde{a}_i$ instead of $a_i \in \{0,1\}$, with independent errors $\Pr(\tilde{a}_i = 1 \mid a_i = 0) = \varphi_0$ and $\Pr(\tilde{a}_i = 0 \mid a_i = 1) = \varphi_1$ that do not depend on y_i or on the design, and $(1-\varphi_0)N_H + \varphi_1 N_A > 0$. For fixed (η, λ), $\mathbb{E}[\hat{N}_\lambda]/\mathbb{E}[\hat{D}_\lambda] = \tilde{\mu}$ with*

$$\tilde{\mu} - \mu_H = \frac{\varphi_1\,\varepsilon\,(\mu_A - \mu_H)}{(1-\varphi_0)(1-\varepsilon) + \varphi_1 \varepsilon}. \tag{9}$$

Proof. $\mathbb{E}[1 - \tilde{a}_i] = (1-\varphi_0)h_i + \varphi_1 a_i$. Repeating part (1) of Theorem 1 with this in place of h_i gives $\mathbb{E}[\hat{N}_\lambda] = (1-\varphi_0)N_H\mu_H + \varphi_1 N_A \mu_A$ and $\mathbb{E}[\hat{D}_\lambda] = (1-\varphi_0)N_H + \varphi_1 N_A$. □

Only machine records passed as human (φ_1) bias the estimate; humans wrongly rejected (φ_0) cost precision, not validity. The bias is at most $\varphi_1 \frac{\varepsilon}{1-\varepsilon}|\mu_A - \mu_H|/(1-\varphi_0)$, a bound an analyst can evaluate from an upper bound on φ_1.

Definition 2 (Regret)**.** A *cell* is a corpus, detector, contamination rate, audit budget and outcome. For an estimation rule r let $\mathrm{sRMSE}_r(c)$ be its root mean squared error in cell c over replications, divided by the standard deviation of the human outcome. The regret of r in c is $\mathrm{sRMSE}_r(c) - \min_{r' \in \mathcal{R}} \mathrm{sRMSE}_{r'}(c)$, where $\mathcal{R}$ is the set of single estimators (naive, deletion, two-phase Hájek, flag-PPI, RC-PPI++). The comparator is the best single estimator in hindsight, which no analyst can select without the truth; regret is non-negative for single estimators and may be negative for a rule that switches between them.

Remark 2 (Selection policy)**.** The policy reports the candidate (naive, deletion, or the audit-based anchor) with the smallest estimated mean squared error. For naive and deletion the error is their bias b_k on the response set, estimated by $\hat{b}_k = \hat{\mu}_k - \hat{\mu}$; since $\mathbb{E}[\hat{b}_k^2] \approx b_k^2 + \mathrm{Var}(\hat{\mu})$, the plug-in $\hat{b}_k^2$ overstates the squared bias and the policy uses $\max(\hat{b}_k^2 - \widehat{\mathrm{se}}^{\,2}, 0)$. The interval reported with the policy is the anchor's.

Numerical checks

Statement	Test (repository `tests/theory/`)
Lemma 1, Proposition 1	`test_identities.py` (exact to 10^{-10})
Theorem 1	`test_exact_statements.py` (representation exact; unbiasedness by enumeration)
Theorem 2	`test_design_based.py` (coverage by simulation)
Corollary 1, Proposition 3	`test_exact_statements.py`
Remark 1, Lemma 2	`test_exact_statements.py` (enumeration)
Propositions 2, 4, 5	`test_design_based.py`, `test_exact_statements.py`